

Grace Chang

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RESEARCH EXPERIENCE

- Expressive Navigation — MIT Media Lab — Fluid Interfaces Group**

Oct 2021 – Mar 2022

 - Prototyped and engineered an affect-aware iOS system leveraging Apple's Vision framework and UIKit to infer subtle emotional cues from facial expressions, translating real-time affective signals into actionable feedback to support clinician-patient therapeutic interactions.
- Jibo Storyteller Persona — MIT Media Lab — Personal Robots Group**

Mar 2022 – Jun 2022

 - Designed and developed a new human-robot interaction persona for Jibo, integrating expressive behaviors, haptic feedback, and backend behavior engineering to create an interactive storytelling companion that strengthened parent-child reading interactions.
- Sonic Stitches — MIT Media Lab — Tangible Media Group**

Sep 2022 – May 2023

 - Conceptualized, fabricated, and engineered a textile-based telepresence system integrating conductive embroidery and Arduino capacitive sensing to capture and transmit touch interactions across distance, addressing the limitations of digital communication by using physical materials to augment remote communication through a greater sense of tactile presence.

WORK EXPERIENCE

- Interface Software Engineer · Microsoft (Experiences & Devices)**

Nov 2024 – Present

 - Led end-to-end interface design and implementation for the Windows Privacy Settings experience serving 4M+ monthly active devices, partnering closely with designers and PMs to translate CSAT telemetry analysis into UI refreshes and architectural decisions that lifted feature discoverability by 10% and cut render latency by 1.57s.
 - Pioneered an LLM-powered developer experience platform that transformed engineering knowledge into interactive AI workflows through MCP tool orchestration, reducing new-hire onboarding friction by 87.5%.
 - Designed intelligent debugging experiences combining telemetry, automated reasoning, and agentic workflows to diagnose regressions across 100+ Windows Settings experiences serving 50K daily retail interactions.
- Distributed Software Engineer · Microsoft (Azure Core)**

Aug 2023 – Nov 2024

 - Optimized distributed infrastructure serving 40B+ daily requests, improving P99.9 latency by up to 250% through architectural improvements and synchronization optimization.
 - Designed adaptive workload management for Azure Resource Manager supporting 20K VM admissions/minute and helping scale the service to 700K requests/hour through a staged, feature-flag-gated rollout across dogfood, canary, and worldwide traffic tiers.
 - Isolated Cosmos DB containers by query pattern, cutting average CPU usage from 70% to 3% and reducing resource query duration from 330ms to 5ms.

PROJECTS

- AI Temperature Monitor — ML-Driven Intelligent Sensing System**
 - Engineered an AI-powered temperature monitoring system integrating Arduino sensors, TensorFlow forecasting, and automated control to demonstrate predictive modeling and intelligent system design for organization-wide technical training.
- Magdalene X² — Multi-Agent Health Intelligence iOS App**
 - Architected and shipped an autonomous AI health companion iOS app to bridge fragmented healthcare experiences by unifying personal health signals across labs, cycle, nutrition, fitness, and sleep, leveraging multi-agent workflows and iterative product design to deliver personalized health insights.
- Dispatcher Voice Agent & Workflow Automation — Freight Dispatch & Carrier Network Platform**
 - Built event-driven AI voice agents triggered by inbound calls and webhook events to automate carrier-to-load coordination for a network of ~2,000 active carriers moving ~500 loads/week, deployed via Docker to Azure Container Apps and launched end-to-end into production.
- Residential Property Servicing Scheduler — Automated Vendor Coordination Service**
 - Diagnosed a production incident causing dropped webhooks, resolved it with backoff-and-jitter retry logic, and built a daily cron-driven pipeline automating a ~\$80K/month vendor-scheduling program with live calendar matching and failure-mode safeguards.

SKILLS

iOS & Apple Platforms	Swift, UIKit, Vision, iOS App Development, Swift
HCI & Prototyping	Tangible User Interfaces, Interaction Design, Physical Computing, Multimodal Interfaces, Figma, Unity, TouchDesigner, Arduino
AI & Intelligent Systems	LLMs, Agentic Systems, MCP / Tool Orchestration, Prompt Engineering, Tool Calling, TensorFlow
Software Engineering	Python, C++, C#, TypeScript, JavaScript, React, Vue, Node.js, Flask, Git, HTML, CSS
Systems	Azure, Distributed Systems, Docker, Kubernetes, Cosmos DB, MongoDB, MySQL

EDUCATION

- Bachelor of Arts in Computer Science (Dual Enrollment)**

May 2023

Massachusetts Institute of Technology · Cambridge, MA

Tangible User Interface · Artificial Intelligence · Software Construction · Software Studio · Data Structures Algorithms
- Wellesley College · Wellesley, MA**

Human-Computer Interaction · Databases with Web Interface